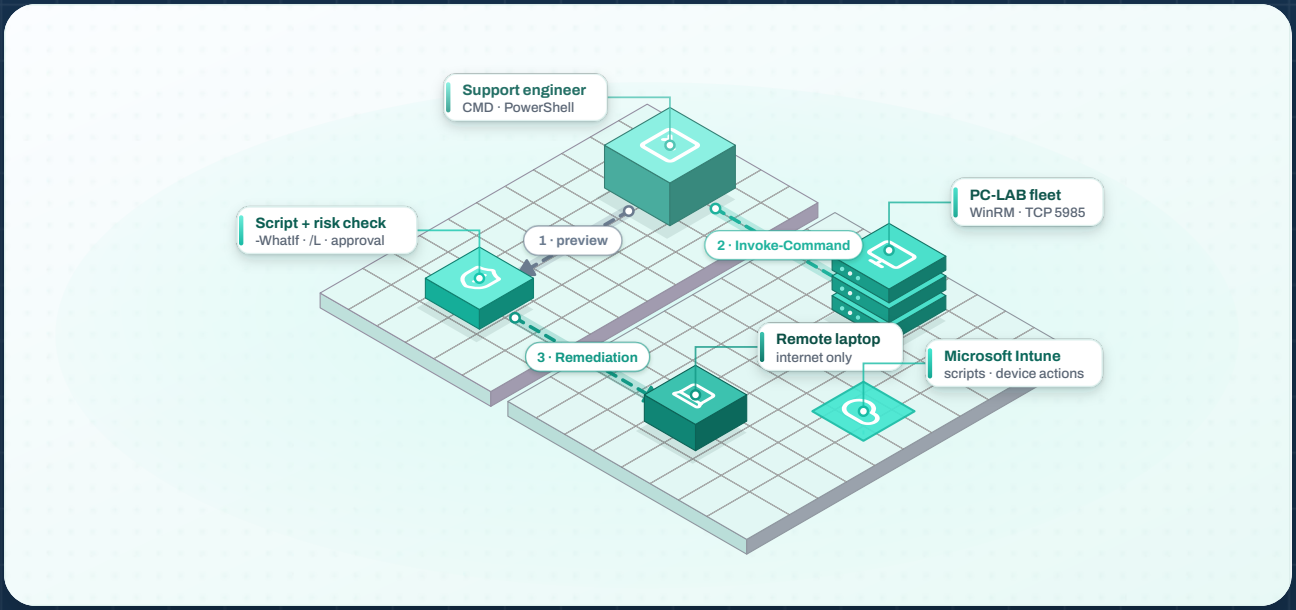
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-WIN-004

The Windows Command-Line Toolkit for Support Engineers

A textbook reference to the everyday Windows support commands, each shown in Command Prompt and PowerShell form, then run remotely with Invoke-Command and at scale with Intune Remediations.

Windows Desktop Support | Command Line and PowerShell | Reference



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Sanitized reference edition — client identifiers removed

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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain the difference between Command Prompt and PowerShell, and when each is the better tool.
- Classify any support command as safe, state-changing or destructive, and apply the matching precautions.
- Follow a fixed order of operations for network diagnosis (ipconfig, ping, nslookup, tracert, port test) and for OS repair (DISM, SFC, chkdsk).
- Use the CMD and PowerShell equivalents for system information, cleanup, networking, accounts, processes, files and logs.
- Create local accounts without exposing passwords, using prompts and SecureString instead of plain text.
- Run the same commands remotely with PowerShell remoting and across a fleet with Intune.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
Reference; each task 1–30 min	Beginner to intermediate	Standard user (read) · local admin (change)	Varies by command; see risk pills

CHAPTER 1

Concepts You Need First

Graphical consoles are fine for one machine and one problem. Support engineers live at the command line because it is faster, it is exact (a command either ran or it didn't), its output can be pasted into a ticket, and the same command can be run on one PC or a thousand. This SOP is a textbook reference to the commands used every day on Windows 10 and 11: what each one does, its Command Prompt (CMD) and PowerShell forms, when to reach for it, and how dangerous it is.

1.1 Two shells, two philosophies

Command Prompt (`cmd.exe`) is the descendant of MS-DOS. Its commands are mostly separate programs (`ipconfig.exe`, `ping.exe`) that print *text*. **PowerShell** is an automation shell built on .NET. Its commands, called *cmdlets*, follow a `Verb-Noun` pattern (`Get-Service`, `Restart-Computer`) and emit *objects* with typed properties.

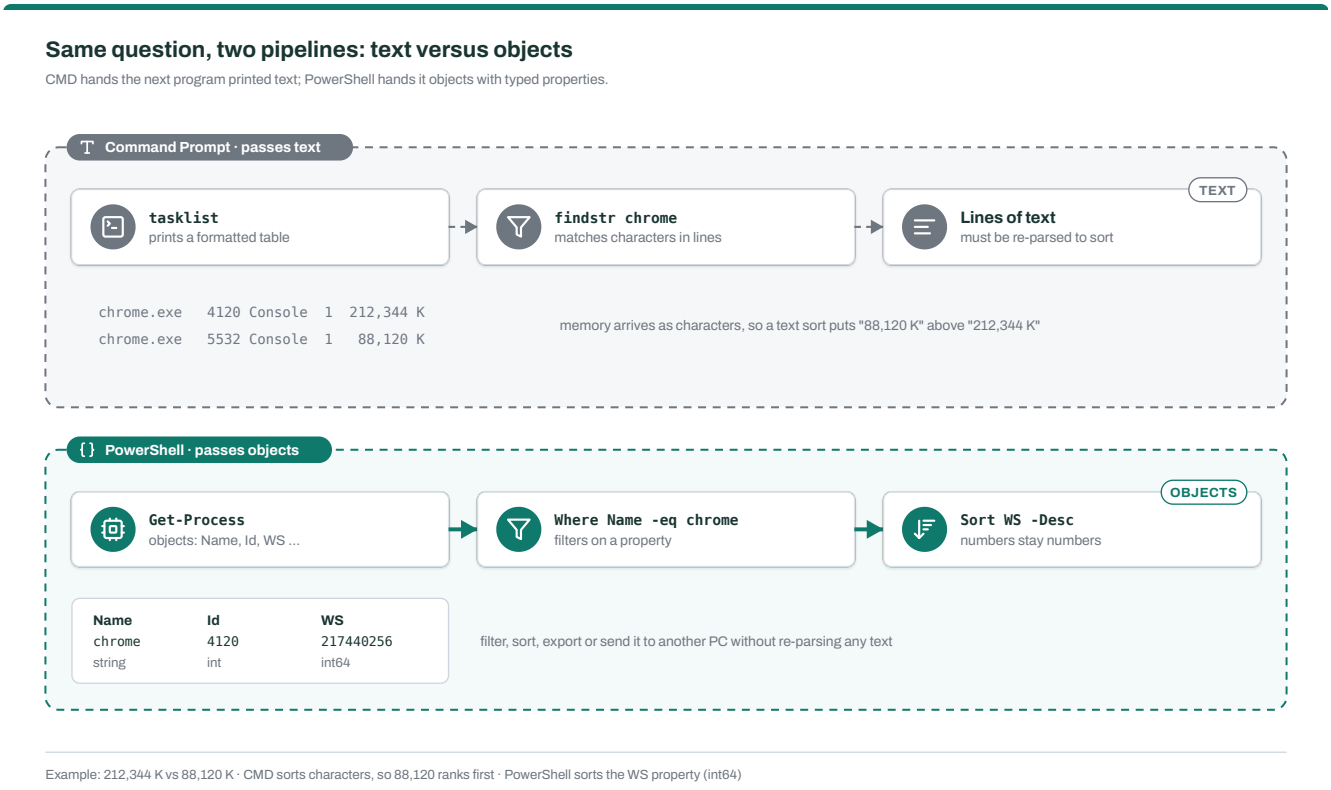


Figure 1. Prefer PowerShell when you need to sort, filter or export, because objects keep numbers as numbers.

KEY CONCEPT · TEXT VERSUS OBJECTS

Asking CMD "which processes use the most memory?" means cutting columns out of a printed table. Asking PowerShell is `Get-Process | Sort-Object WS -Descending | Select-Object -First 5`, because `WS` (working set) is a number, not characters. Every classic CMD tool still runs inside PowerShell, so you never lose anything by working in PowerShell; you gain filtering, error handling and remoting.

	Command Prompt	PowerShell
Output	Text	Objects (format with `Format-Table`, export with `Export-Csv`, `ConvertTo-Json`)
Help	`command /?`	`Get-Help Get-Service -Examples`
Last result code	`%ERRORLEVEL%`	`$?` (success) and `$LASTEXITCODE` (native programs)
Preview a change	Rarely available (`robocopy /L`)	`-WhatIf` and `-Confirm` on most changing cmdlets
Remote execution	Some tools take `/m \\PC` or `\\PC`	`Invoke-Command`, `Enter-PSSession` for anything
Gotcha	Variables are `%NAME%`; `%` doubles in batch files	Some aliases shadow programs: `sc` means `Set-Content`; type `sc.exe`

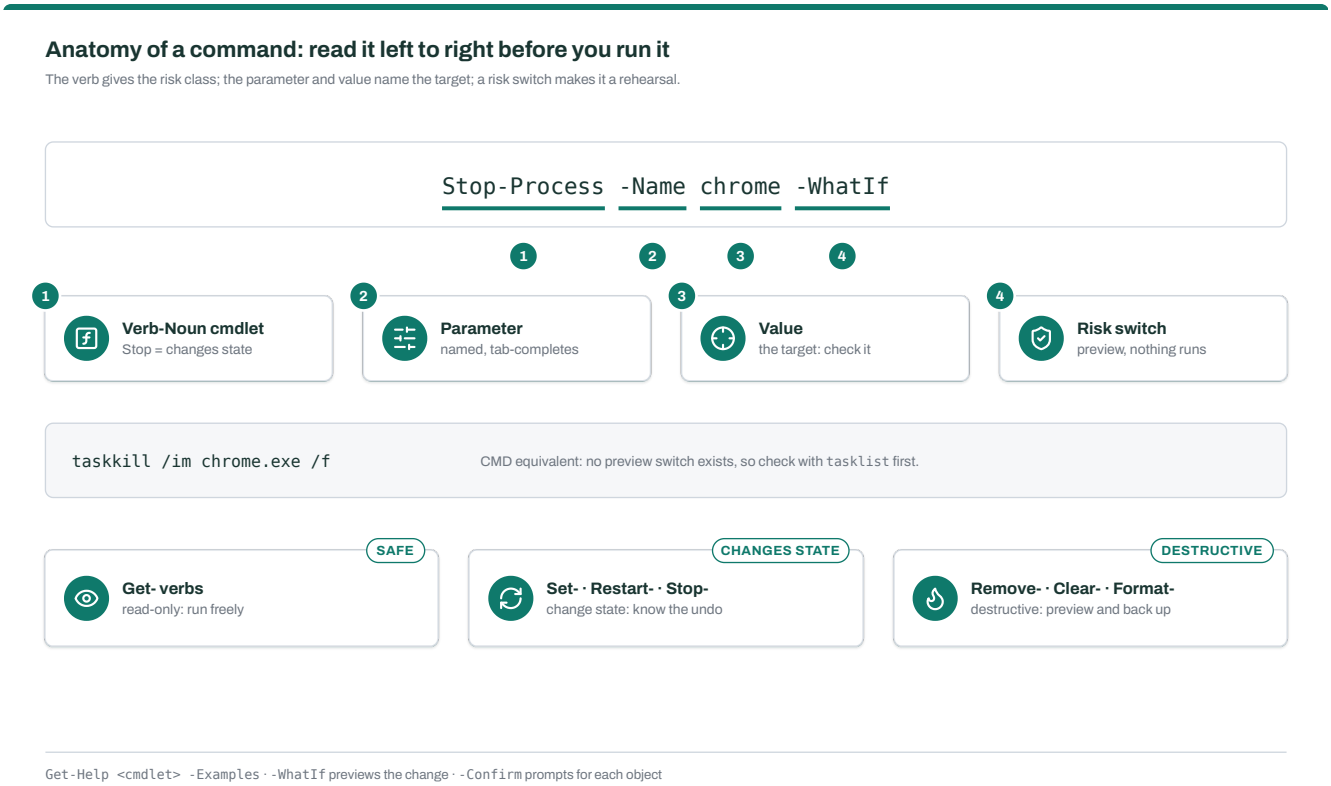


Figure 2. Check the target before pressing Enter; CMD tools rarely offer a preview, so run the matching read-only command first.

1.2 Elevation: why "Access is denied"

With User Account Control (UAC), even an administrator runs programs with a *standard* token. Commands that change the system (services, drivers, system files, other users' processes) need an *elevated* shell: right-click **Terminal**, **Command Prompt** or **PowerShell** › **Run as administrator**. The title bar then begins with *Administrator:*. In PowerShell you can check with `([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole('Administrators')`. Stay unelevated for anything user-specific, such as mapping drives, because elevated and standard sessions do not share drive mappings.

1.3 Safe, state-changing and destructive commands

Every command in this SOP carries a risk pill. Learn to classify commands before you type them:

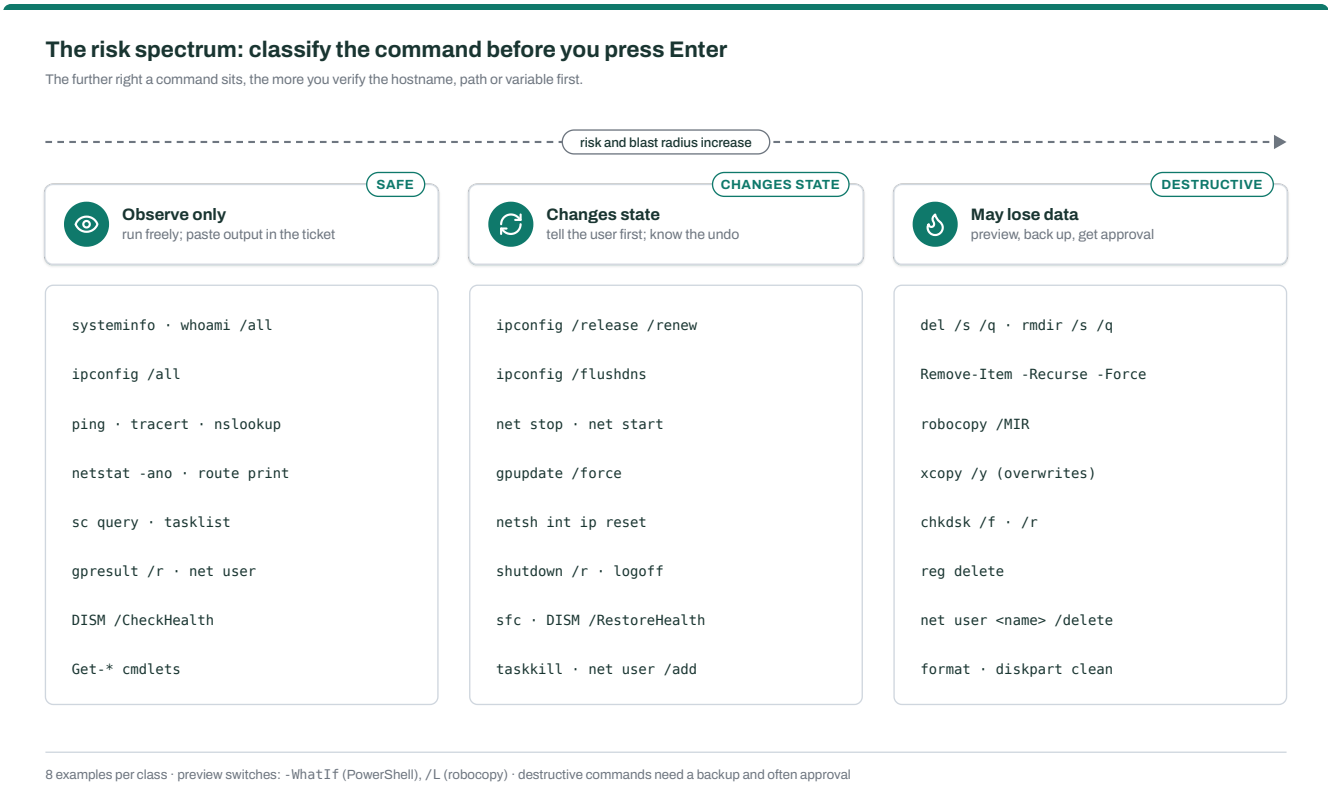


Figure 3. Read-only commands can be run freely; destructive ones need a preview, a backup and often approval.

Pill	Meaning and required precaution
SAFE	Reads state only. Run without asking; save the output to the ticket.
CHANGES STATE	Changes configuration or interrupts the user (network drop, service restart, reboot). Warn the user, know the undo.
DESTRUCTIVE	Deletes or overwrites data, or takes a disk offline for hours. Verify the exact target, preview, back up, and follow change control.

RULE · PATHS ARE THE DANGER, NOT THE COMMAND

`rmdir /s /q` is harmless on `C:\Temp\old-logs` and catastrophic on `C:\Users`. A misplaced space or an empty variable (`Remove-Item "$folder\*"` when `$folder` is empty becomes `\*`, the root of the current drive) turns a clean-up into an outage. Always echo or `-WhatIf` destructive commands first.

1.4 Choosing a method

A · Command Prompt

Built-in tools, available everywhere including WinRE.

BEST FOR: QUICK CHECKS, RECOVERY SHELLS

B · PowerShell

Objects, filtering, -WhatIf, scripting and evidence.

BEST FOR: ANYTHING REPEATABLE

C · Remote PowerShell

Invoke-Command and Enter-PSSession over WinRM.

BEST FOR: ONE TO HUNDREDS OF PCS ON THE NETWORK

D · Intune

Remediations, platform scripts and device actions.

BEST FOR: INTERNET-ONLY AND FLEET-WIDE WORK

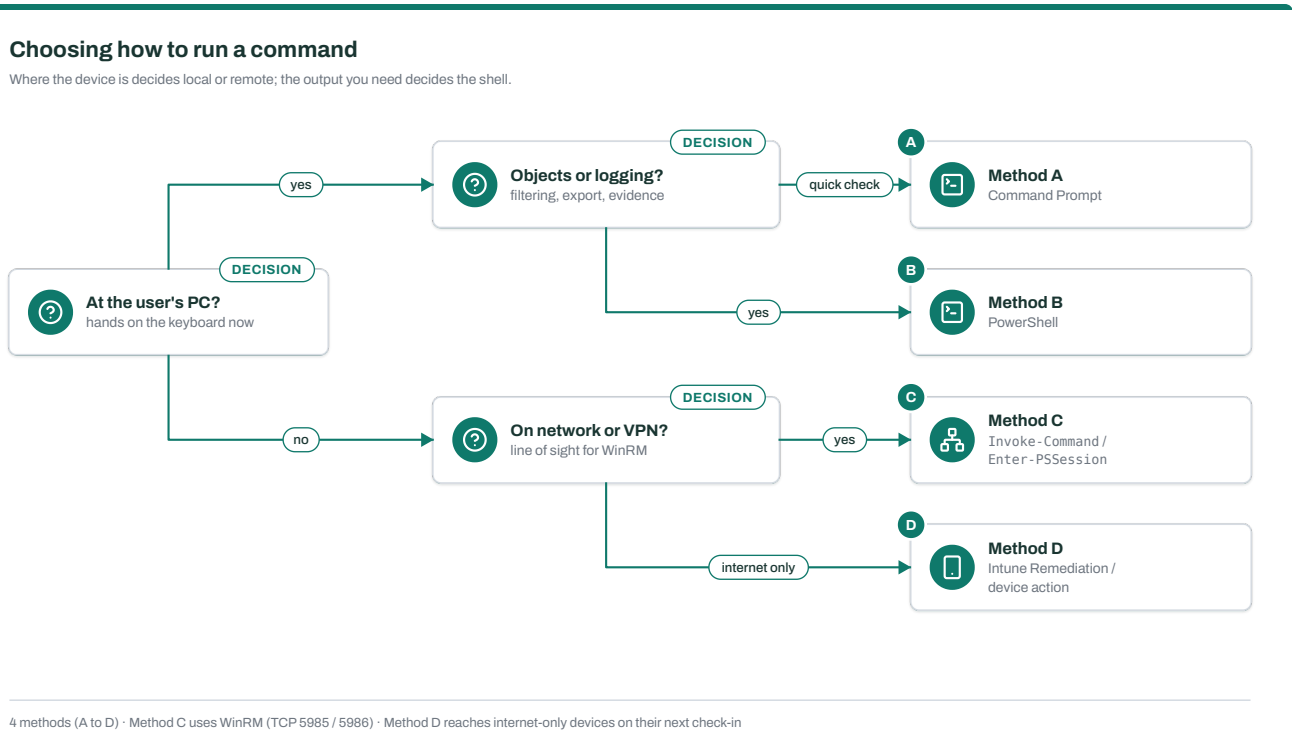


Figure 4. Pick the shell by the output you need and the remote path by whether the device is on the corporate network.

CHAPTER 2

Access, Prerequisites and Working Habits

2.1 Level of access required

Command family	Minimum access	Notes
Information, network diagnosis, `gpresult /r`, `Get-*`	Standard user	Some details (other users' processes, Security log) need elevation.
Services, drivers, DISM, SFC, chkdsk, `netsh` resets	Local administrator, elevated	Use the device's Windows LAPS password, not a shared account.
Local users and groups	Local administrator	Creating admins should itself be a tracked change.
Domain users and groups (`net user /domain`, AD cmdlets)	Delegated rights in AD	Read is open to authenticated users; changes need delegation.
Remote PowerShell	Administrators (or Remote Management Users) on the target; WinRM enabled	JEA (Just Enough Administration) endpoints can expose only approved commands.
Intune scripts, Remediations, device actions	Intune role with *Device configurations* and *Remote tasks* permissions	Help Desk Operator covers many device actions; Remediations need appropriate licensing.

2.2 Prerequisites and habits

- [ ] A ticket or change record, and the user's consent before anything that interrupts their session.
- [ ] Capture evidence: `command > %PUBLIC%\Documents\out.txt` in CMD, or `Start-Transcript` … `Stop-Transcript` in PowerShell.
- [ ] Confirm the target: `hostname` and `whoami` before any change, especially in remote sessions.
- [ ] Use `-WhatIf`, `robocopy /L` or an `echo` of the expanded command before destructive operations.
- [ ] Never type passwords on a command line; use prompts, `Get-Credential` or `Read-Host -AsSecureString`.

CHAPTER 3

Methods A and B · Information, Cleanup and Networking

Each family below gives a short explanation of the concept, then a reference table with the CMD form, the PowerShell equivalent and the risk. Commands are Method A (CMD) or Method B (PowerShell); both run locally.

3.1 System information

Start every ticket by establishing *what* you are working on: OS version and build, uptime, model, logged-on user, domain. Half of all "weird" issues are explained by an old build, a pending reboot or a machine that has not restarted in 60 days.

Task	CMD	PowerShell	Risk
OS, build, uptime, hotfixes	`systeminfo`	`Get-ComputerInfo -Property OsName, OsBuildNumber, OsLastBootUpTime`	SAFE
Computer and user	`hostname` · `whoami /all`	`$env:COMPUTERNAME` · `whoami /groups`	SAFE
Version dialog	`winver`	`[Environment]::OSVersion`	SAFE
Hardware, model, serial	`msinfo32`	`Get-CimInstance Win32_ComputerSystem` · `Win32_BIOS`	SAFE
Disk space	`fsutil volume diskfree C:`	`Get-Volume C` · `Get-PSDrive C`	SAFE

> **NOTE · WMIC IS RETIRED**
>
> Older guides use `wmic`. It is deprecated and no longer installed by default on recent Windows 11 builds. Use `Get-CimInstance`, which queries the same WMI classes.

3.2 Cleanup

Temporary files accumulate in each user's `%TEMP%` and in `C:\Windows\Temp`. Files in use are locked and are simply skipped, so a temp cleanup is safe for the user's data but can break an installer that is running right now. The component store (WinSxS) is cleaned with DISM, never by deleting files by hand.

Task	CMD	PowerShell	Risk
Clear the user's temp	`del /s /q %temp%\*`	`Remove-Item "$env:TEMP\*" -Recurse -Force -ErrorAction SilentlyContinue`	CHANGES
Disk Cleanup	`cleanmgr` · `cleanmgr /sageset:1` then `/sagerun:1`	Settings › System › Storage › Storage Sense	CHANGES
Shrink WinSxS	`DISM /Online /Cleanup-Image /StartComponentCleanup`	`Repair-WindowsImage -Online -StartComponentCleanup`	CHANGES
Find the largest files	`dir /s /o-s %USERPROFILE%`	`Get-ChildItem $HOME -Recurse -File | Sort-Object Length -Desc | Select-Object -First 20`	SAFE

WATCH OUT · NEVER POINT A RECURSIVE DELETE AT A VARIABLE YOU HAVE NOT CHECKED

`del /s /q %temp%\*` is safe because `%TEMP%` is always defined in an interactive session. In a script running as SYSTEM, `%TEMP%` is `C:\Windows\Temp`, not the user's folder. Test `echo %temp%` first.

3.3 Networking

Network problems are diagnosed bottom-up: local configuration, local segment, name resolution, routing, then the application port. Each step either proves a layer works or tells you where to stop. Do not jump to `netsh` resets before you know which layer failed.

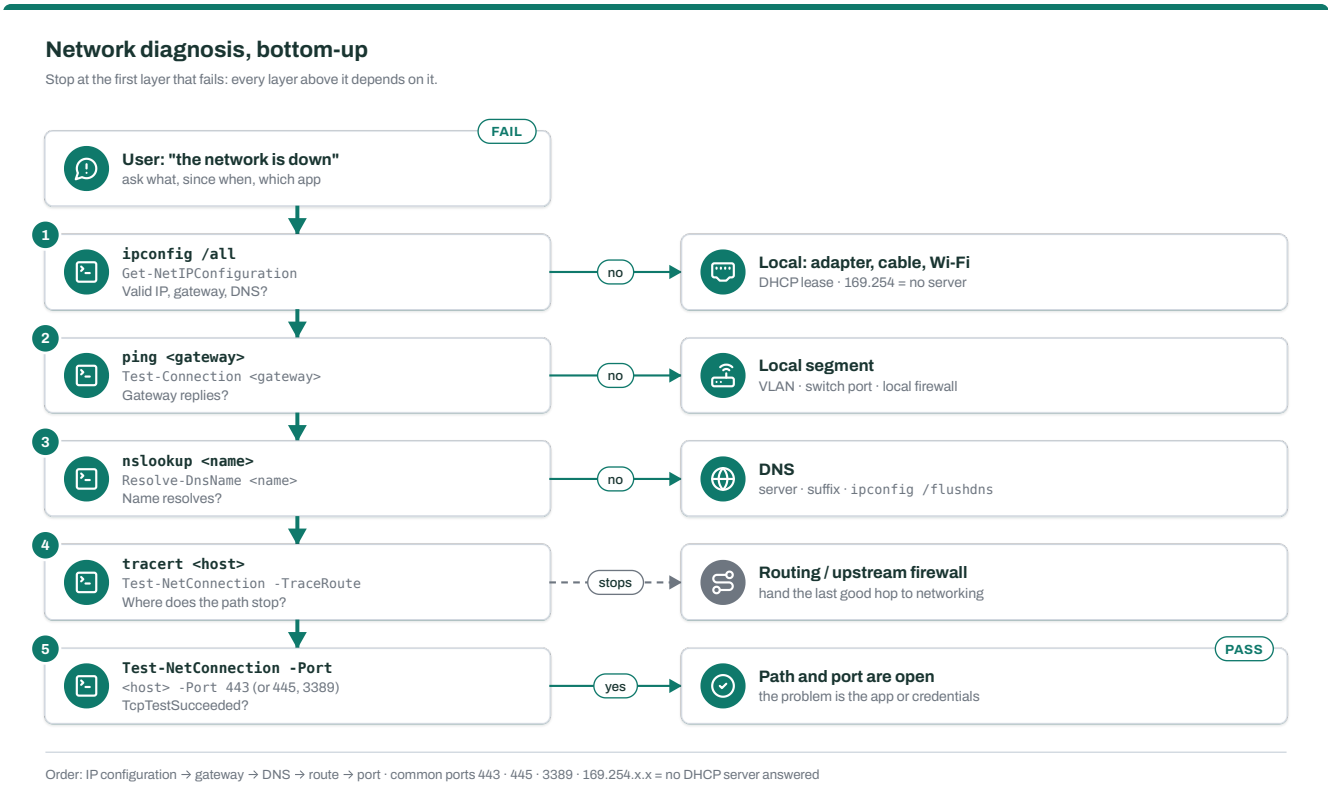


Figure 5. Work up from the local configuration and stop at the first failing layer, because everything above it depends on it.

Task	CMD	PowerShell	Risk
IP configuration	`ipconfig /all`	`Get-NetIPConfiguration -Detailed`	SAFE
Reachability	`ping 10.0.0.1` · `ping -n 20 host`	`Test-Connection 10.0.0.1 -Count 4`	SAFE
Name resolution	`nslookup host` · `nslookup host 10.0.0.10`	`Resolve-DnsName host -Server 10.0.0.10`	SAFE
Route and per-hop loss	`tracert host` · `pathping host`	`Test-NetConnection host -TraceRoute`	SAFE
TCP port open?	(none built in)	`Test-NetConnection host -Port 443`	SAFE
Listening ports and owners	`netstat -ano`	`Get-NetTCPConnection -State Listen`	SAFE
Neighbours and routes	`arp -a` · `route print`	`Get-NetNeighbor` · `Get-NetRoute`	SAFE
Flush DNS cache	`ipconfig /flushdns`	`Clear-DnsClientCache`	CHANGES

Release and renew DHCP	`ipconfig /release` · `/renew`	`Restart-NetAdapter -Name Ethernet`	CHANGES
Reset TCP/IP and Winsock	`netsh int ip reset` · `netsh winsock reset`	(same commands; restart required)	CHANGES

WORKED EXAMPLE · READING THE ORDER OF OPERATIONS

A user cannot open an intranet site. `ipconfig /all` shows a valid lease. `ping` to the gateway works. `nslookup intranet.contoso.com` returns *Non-existent domain*. Stop: the problem is DNS, so there is no point running `tracert` or resetting Winsock. Check which DNS servers the client uses and whether the record exists.

WATCH OUT · RELEASING YOUR OWN CONNECTION

`ipconfig /release` in a remote session disconnects you, and the renew never arrives. Do it locally where possible; remotely, chain both on one line (`ipconfig /release && ipconfig /renew`) and expect to reconnect. A `netsh` reset needs a restart and removes static IP settings: record them first.

CHAPTER 4

Methods A and B · Accounts, Sessions, Processes and Services

4.1 Local users and groups

Local accounts exist only on one PC and are stored in its Security Account Manager (SAM) database. On managed fleets the only local account that should normally be used is the LAPS-managed administrator; everything else is a domain or Entra ID identity. When you must create a local account (a kiosk, a lab instrument, a break-glass test), never put its password on the command line: it ends up in console history, in process-creation logs and in EDR telemetry.

Task	CMD	PowerShell	Risk
List users / one user	`net user` · `net user kiosk01`	`Get-LocalUser`	SAFE
List group members	`net localgroup Administrators`	`Get-LocalGroupMember Administrators`	SAFE
Create user (prompted password)	`net user kiosk01 * /add`	`New-LocalUser kiosk01 -Password (Read-Host -AsSecureString)`	CHANGES
Add to a group	`net localgroup "Remote Desktop Users" kiosk01 /add`	`Add-LocalGroupMember -Group 'Remote Desktop Users' -Member kiosk01`	CHANGES
Disable / delete	`net user kiosk01 /active:no` · `/delete`	`Disable-LocalUser kiosk01` · `Remove-LocalUser`	DESTRUCTIVE

The asterisk in `net user kiosk01 * /add` tells Windows to *prompt* for the password without echoing it. The PowerShell version receives a `SecureString`:

CREATE A LOCAL ACCOUNT WITHOUT EXPOSING THE PASSWORD (ELEVATED) — POWERSHELL

```
$pw = Read-Host -Prompt 'Password for kiosk01' -AsSecureString
New-LocalUser -Name 'kiosk01' -Password $pw -FullName 'Lab kiosk' `
  -Description "Created for CHG0012345" -PasswordNeverExpires:$false
Add-LocalGroupMember -Group 'Users' -Member 'kiosk01'
Remove-Variable pw
```

CAUTION · ADDING TO ADMINISTRATORS

The original quick reference added new users to *Administrators* by default. Do that only with an approved reason, prefer the least-privileged group that works (Users, Remote Desktop Users, a specific application group), and remove the membership when the need ends.

4.2 Sessions, restarts and consoles

Task	CMD	PowerShell	Risk
Who is signed in	`quser` (`query user`)	`quser`	SAFE
Restart with warning	`shutdown /r /t 300 /c "Restart for CHG0012345"`	`Restart-Computer -Confirm`	CHANGES
Restart now	`shutdown /r /t 0`	`Restart-Computer -Force`	CHANGES
Shut down now / abort	`shutdown /s /t 0` · `shutdown /a`	`Stop-Computer`	CHANGES
Sign out a session	`logoff` · `logoff <sessionID>`	(same)	CHANGES
Open consoles	`devmgmt.msc` · `services.msc` · `eventvwr.msc` · `compmgmt.msc` · `diskmgmt.msc`	(same)	SAFE
Registry	`regedit` · `reg query` · `reg export`	`Get-ItemProperty HKLM:\…`	SAFE to read
Registry changes	`reg add` · `reg delete`	`Set-ItemProperty` · `Remove-ItemProperty`	DESTRUCTIVE

GOOD PRACTICE · EXPORT BEFORE YOU EDIT

`reg export "HKLM\SOFTWARE\Contoso\App" %PUBLIC%\Documents\app-before.reg` gives you a one-command rollback (`reg import`). `shutdown /r /t 0` gives a user no chance to save work: use a delay and `/c` comment unless the PC is already unusable.

4.3 Processes and services

A **process** is a running program; a **service** is a program the Service Control Manager runs in the background, often before anyone signs in. Stopping a process loses its unsaved data; stopping a service can break features for every user of the PC.

Task	CMD	PowerShell	Risk
List processes	`tasklist` · `tasklist /svc`	`Get-Process | Sort-Object CPU -Desc`	SAFE
End a process	`taskkill /PID 4321` · `/IM app.exe /F`	`Stop-Process -Id 4321`	CHANGES
Service state and config	`sc query Spooler` · `sc qc Spooler`	`Get-Service Spooler | Select *`	SAFE
Start / stop / restart	`net stop Spooler` · `net start Spooler`	`Restart-Service Spooler`	CHANGES
Start type	`sc config Spooler start= auto`	`Set-Service Spooler -StartupType Automatic`	CHANGES

CHAPTER 5

Methods A and B · Files, Repair, Policy and Logs

5.1 File operations

`xcopy` still works, but **robocopy** ("robust copy") is the modern tool: it retries on network errors, preserves timestamps and attributes, logs every file and can list what it *would* do with `/L`. Its `/MIR` switch makes the destination an exact mirror, which means it *deletes* anything in the destination that is not in the source.

Task	CMD	PowerShell	Risk
Copy a tree (legacy)	`xcopy C:\source D:\backup /s /e /y`	`Copy-Item C:\source D:\backup -Recurse`	OVERWRITES
Copy a tree (preferred)	`robocopy C:\source D:\backup /E /COPY:DAT /R:2 /W:5 /LOG:C:\Temp\rc.log`	(call robocopy)	OVERWRITES
Preview a mirror	`robocopy C:\src D:\dst /MIR /L`	—	SAFE
Mirror	`robocopy C:\src D:\dst /MIR`	—	DESTRUCTIVE
Delete a file	`del /f /q C:\path\file.txt`	`Remove-Item C:\path\file.txt -WhatIf`	DESTRUCTIVE
Delete a folder tree	`rmdir /s /q C:\path\folder`	`Remove-Item C:\path\folder -Recurse -WhatIf`	DESTRUCTIVE

WATCH OUT · /MIR WITH THE ARGUMENTS REVERSED

Swap source and destination in a `/MIR` command and robocopy faithfully deletes the user's data to match an empty backup folder. Always run the same command with `/L` first and read the summary counts.

5.2 System repair: DISM, SFC and chkdsk

Three tools repair three different layers. **DISM** repairs the *component store* (`C:\Windows\WinSxS`), the master copy of Windows files. **SFC** repairs the *live system files* by copying from that store. **chkdsk** repairs the *file system* (NTFS metadata) underneath both. Run them in that order: a broken store makes SFC fail, and SFC cannot fix a damaged file system.

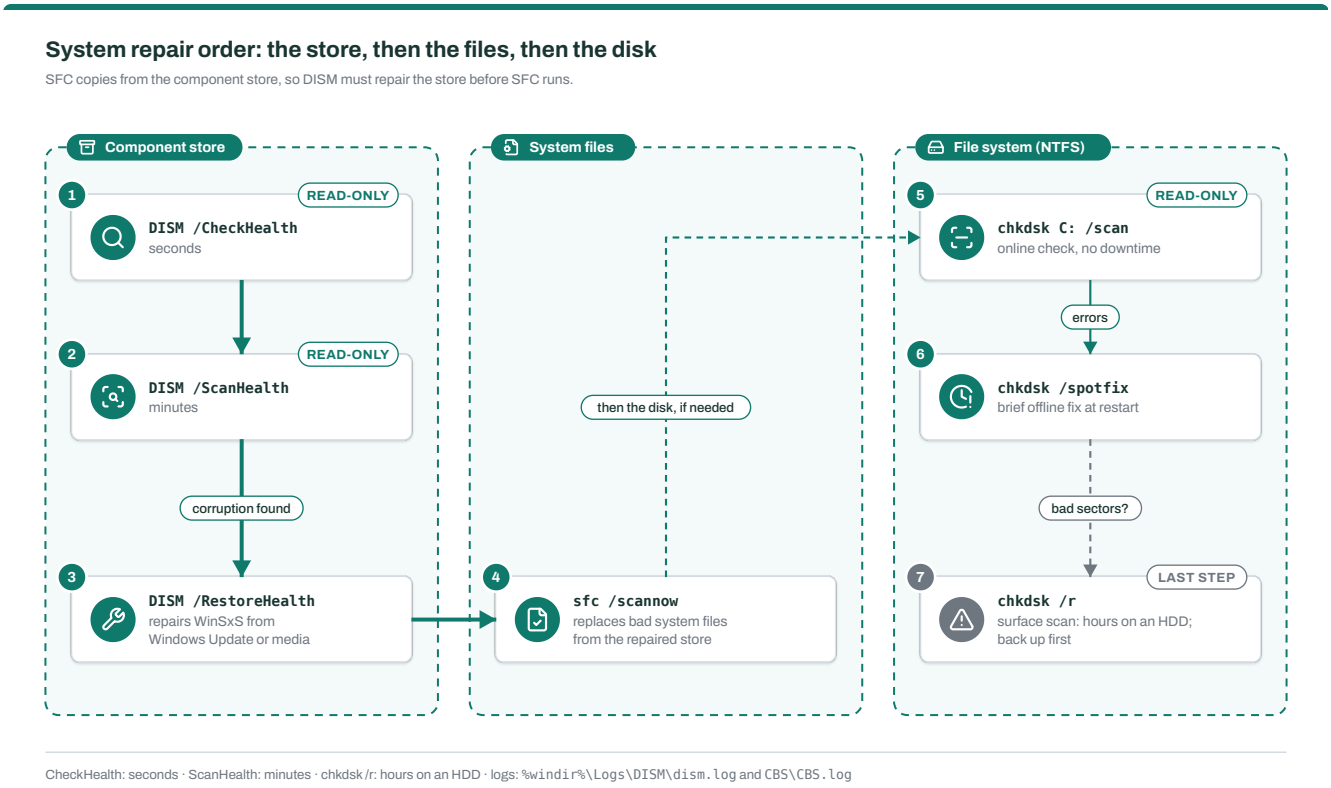


Figure 6. Repair the component store before running SFC; the surface scan is a last step for suspected hardware faults.

Task	CMD (elevated)	PowerShell (elevated)	Risk
Quick store check	`DISM /Online /Cleanup-Image /CheckHealth`	`Repair-WindowsImage -Online -CheckHealth`	SAFE
Full store scan	`DISM /Online /Cleanup-Image /ScanHealth`	`Repair-WindowsImage -Online -ScanHealth`	SAFE
Repair store	`DISM /Online /Cleanup-Image /RestoreHealth`	`Repair-WindowsImage -Online -RestoreHealth`	CHANGES
Repair system files	`sfc /scannow` (check only: `sfc /verifyonly`)	(same)	CHANGES
Online volume scan	`chkdsk C: /scan`	`Repair-Volume -DriveLetter C -Scan`	SAFE
Fix found errors	`chkdsk C: /spotfix` · `chkdsk C: /f`	`Repair-Volume -DriveLetter C -SpotFix`	OFFLINE
Surface scan	`chkdsk C: /r`	`Repair-Volume -DriveLetter C -OfflineScanAndFix`	HOURS OFFLINE

WHY · CHKDSK ON C: ASKS TO "SCHEDULE AT NEXT RESTART"

Fixing a volume requires exclusive access, and Windows is running from C:. `/f` and `/r` on the system drive are therefore queued for the next boot. `/r` reads every sector and can take hours on a large disk; on SSDs it adds little, and on a failing HDD it can accelerate failure. Back up first, and prefer `/scan` plus `/spotfix` on healthy hardware.

NOTE · DISM ERROR 0X800F081F

"The source files could not be found" means DISM could not download repair files. Point it at matching install media: `DISM /Online /Cleanup-Image /RestoreHealth /Source:WIM:D:\sources\install.wim:1 /LimitAccess`, choosing the index that matches the installed edition (`DISM /Get-WimInfo`).

5.3 Group Policy and event logs

Task	CMD	PowerShell	Risk
Re-apply policy	`gpupdate /force`	`Invoke-GPUpdate` (RSAT; remote too)	CHANGES
What applied	`gpresult /r` · `gpresult /h C:\Temp\gp.html`	`Get-GPResultantSetOfPolicy` (RSAT)	SAFE
Recent errors	`wevtutil qe System /c:20 /rd:true /f:text`	`Get-WinEvent -FilterHashtable @{LogName='System'; Level=2} -MaxEvents 20`	SAFE
Export a log	`wevtutil epl System C:\Temp\System.evtx`	(same)	SAFE

CHAPTER 6

Methods C and D · Remote and Enterprise Execution

6.1 Method C · PowerShell remoting

PowerShell remoting runs commands on other computers over **WinRM** (Windows Remote Management, the WS-Management protocol). In a domain it authenticates with Kerberos and encrypts the session even on port 5985. It is enabled by default on Windows Server and is commonly enabled on managed clients by Group Policy (`Enable-PSRemoting` locally).

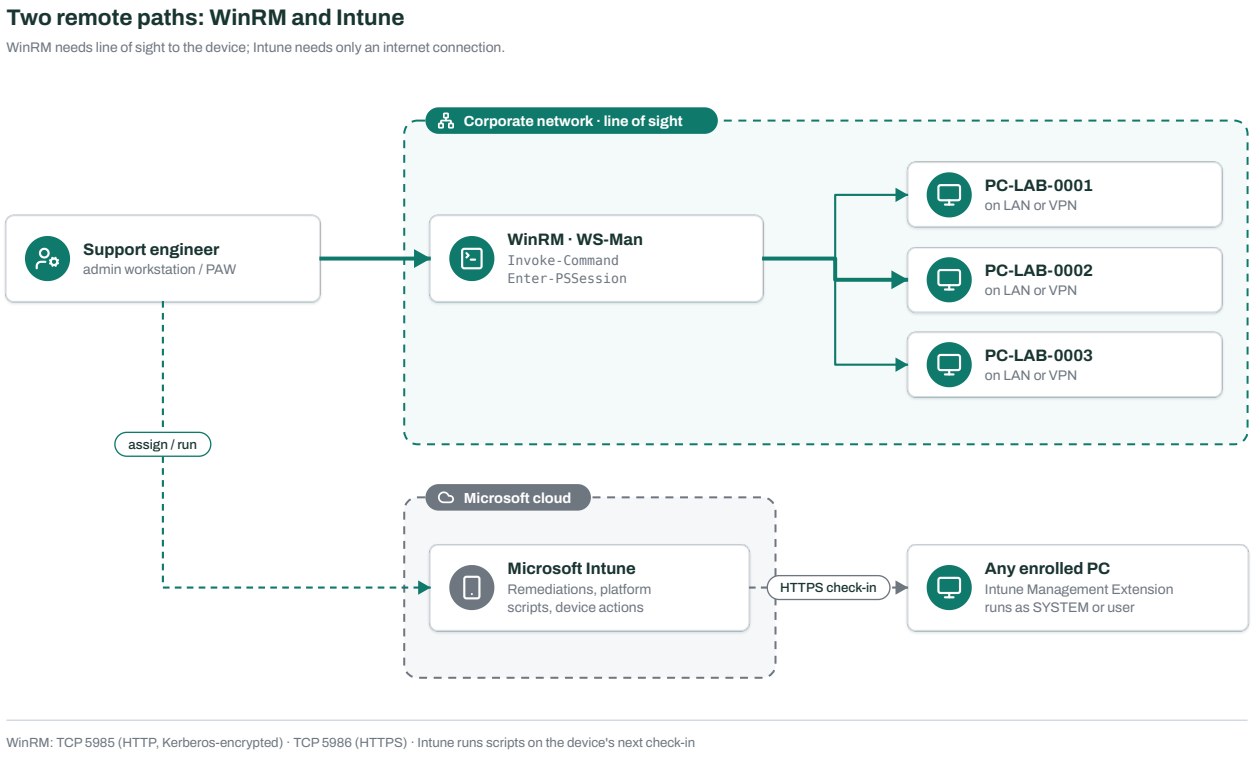


Figure 7. Use WinRM for devices you can reach directly and Intune for devices that only have an internet connection.

INTERACTIVE AND FAN-OUT REMOTING — POWERSHELL

```powershell
#  Interactive: like sitting at the PC (prompt changes to [PC-LAB-0001]: PS>)
Enter-PSSession -ComputerName PC-LAB-0001
Get-Service Spooler; Exit-PSSession

#  Fan-out: same script block on many PCs in parallel, results tagged with PSComputerName
$pcs = 'PC-LAB-0001','PC-LAB-0002','PC-LAB-0003'
Invoke-Command -ComputerName $pcs -ScriptBlock {
  [pscustomobject]@{ Build = (Get-CimInstance Win32_OperatingSystem).BuildNumber
                     LastBoot = (Get-CimInstance Win32_OperatingSystem).LastBootUpTime
                     FreeGB = [math]::Round((Get-Volume C).SizeRemaining / 1GB, 1) }
```

```
} | Select-Object PSComputerName, Build, LastBoot, FreeGB | Export-Csv .\fleet.csv -NoTypeInformation

#  Classic CMD tools that accept a remote target
shutdown /r /m \\PC-LAB-0001 /t 600 /c "Restart for CHG0012345"
sc.exe \\PC-LAB-0001 query Spooler
```

WATCH OUT · THE DOUBLE HOP

Inside a remote session your credentials cannot be forwarded to a third machine, so `Copy-Item \\fs01\share\…` from within `Invoke-Command` fails with access denied. Copy the files to the target first (`Copy-Item -ToSession`), or use a share the computer account can read.

6.2 Base template · support snapshot with evidence

This read-only template collects the facts most tickets need, locally or from remote PCs, and writes a JSON file. Its only optional change, clearing temp files, honours `-WhatIf`.

GET-SUPPORTSNAPSHOT.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Collect a support snapshot (OS, uptime, disk, network, failed services, recent errors) from
one or more PCs.
.NOTES     Read-only unless -ClearTemp is used. Remote targets need WinRM and admin rights.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [string[]] $ComputerName = $env:COMPUTERNAME,
  [string]   $TestHost     = 'fs01.contoso.com',
  [int]      $TestPort     = 445,
  [switch]   $ClearTemp,
  [string]   $OutFile      = ".\SupportSnapshot-$(Get-Date -f yyyyMMdd-HHmm).json"
)
$ErrorActionPreference = 'Stop'
$collect = {
  param($TestHost, $TestPort)
  $os  = Get-CimInstance Win32_OperatingSystem
  $ip  = Get-NetIPConfiguration | Where-Object IPv4DefaultGateway
  [pscustomobject]@{
    Computer      = $env:COMPUTERNAME
    OS            = "$($os.Caption) $($os.BuildNumber)"
    LastBoot      = $os.LastBootUpTime
    FreeGB_C      = [math]::Round((Get-Volume C).SizeRemaining / 1GB, 1)
    IPv4          = $ip.IPv4Address.IPAddress
    Gateway       = $ip.IPv4DefaultGateway.NextHop
    DnsServers    = ($ip.DNSServer | Where-Object AddressFamily -eq 2).ServerAddresses
    PortTest      = (Test-NetConnection $TestHost -Port $TestPort -WarningAction
SilentlyContinue).TcpTestSucceeded
    StoppedAuto   = (Get-CimInstance Win32_Service -Filter "StartMode='Auto' AND State<>'Running'").Name
    RecentErrors  = Get-WinEvent -FilterHashtable @{ LogName='System'; Level=2; StartTime=(Get-
Date).AddDays(-1) } -MaxEvents 10 -ErrorAction SilentlyContinue |
                    Select-Object TimeCreated, ProviderName, Id
  }
}
$results = foreach ($pc in $ComputerName) {
  if ($pc -eq $env:COMPUTERNAME) { & $collect $TestHost $TestPort }
```

```
  else { Invoke-Command -ComputerName $pc -ScriptBlock $collect -ArgumentList $TestHost, $TestPort }
}
if ($ClearTemp) {
  foreach ($pc in $ComputerName) {
    if ($PSCmdlet.ShouldProcess($pc, 'Clear C:\Windows\Temp')) {
      Invoke-Command -ComputerName $pc -ScriptBlock { Remove-Item 'C:\Windows\Temp\*' -Recurse -Force -
ErrorAction SilentlyContinue }
    }
  }
}
$results | ConvertTo-Json -Depth 4 | Out-File $OutFile -Encoding utf8
$results | Select-Object Computer, OS, LastBoot, FreeGB_C, PortTest | Format-Table -AutoSize
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION — POWERSHELL

```
.\Get-SupportSnapshot.ps1                                          # this PC
.\Get-SupportSnapshot.ps1 -ComputerName PC-LAB-0001,PC-LAB-0002 -TestHost intranet.contoso.com -
TestPort 443
.\Get-SupportSnapshot.ps1 -ComputerName PC-LAB-0001 -ClearTemp -WhatIf
```

6.3 Method D · Intune for devices you cannot reach

Remote and hybrid workers are often not on the corporate network, so WinRM cannot reach them. Intune can, because the device checks in over HTTPS.

- **Device actions** (Devices › All devices › <device>): `Restart`, `Sync`, `Collect diagnostics` (gathers logs and command output into a downloadable package) and others.
- **Remediations**: a detection script and a remediation script, scheduled or run on demand (`Run remediation` on a device). Ideal for "check and fix" tasks such as clearing a full temp folder or restarting a stuck service.
- **Platform scripts**: one-off PowerShell, run once per device or user.

EXAMPLE REMEDIATION PAIR · STUCK PRINT SPOOLER — POWERSHELL

```
#  Detection: exit 1 if the spooler is not running
if ((Get-Service Spooler).Status -ne 'Running') { Write-Output 'Spooler stopped'; exit 1 }
Write-Output 'Spooler running'; exit 0

#  Remediation: clear stuck jobs and restart
Stop-Service Spooler -Force
Remove-Item "$env:SystemRoot\System32\spool\PRINTERS\*" -Force -ErrorAction SilentlyContinue
Start-Service Spooler
Write-Output "Spooler: $((Get-Service Spooler).Status)"; exit 0   # last output line is shown in the
Intune report
```

CHAPTER 7

Verification and Troubleshooting

7.1 Verifying a command did what you think

- [ ] Check the exit status: `echo %ERRORLEVEL%` in CMD (0 usually means success), `$?` or `$LASTEXITCODE` in PowerShell.
- [ ] Re-run the matching read-only command (for `net start`, `sc query`; for a mapping, `net use`).
- [ ] Look in the event log for the time of the change.
- [ ] Ask the user to confirm the symptom is gone, and attach the transcript or output file to the ticket.

7.2 Common errors and what they mean

Message	Cause and fix
Access is denied / requires elevation	Shell not elevated, or the account is not a local admin. Reopen with Run as administrator.
'x' is not recognized as an internal or external command	Typo, tool not installed (RSAT, Sysinternals), or not on `PATH`. In PowerShell, a CMD built-in like `dir /s` must be called as `cmd /c dir /s`.
running scripts is disabled on this system	Execution policy. Use signed scripts or a policy set by GPO/Intune; for a one-off, `powershell -ExecutionPolicy Bypass -File .\x.ps1` where allowed.
WinRM cannot complete the operation	Target offline, WinRM not enabled, firewall blocking 5985, or name not resolving. `Test-WSMan PC-LAB-0001`.
SFC found corrupt files but was unable to fix some	Run `DISM /RestoreHealth` first, then SFC again. Read `%windir%\Logs\CBS\CBS.log`.
DISM 0x800f081f	No repair source. Use `/Source` with matching media and `/LimitAccess`.
Chkdsk cannot run because the volume is in use	Expected for C:. Answer `Y` to schedule at restart, or use `/scan` online.
sc config: "DESCRIPTION: ... USAGE"	Missing space after `=`. The syntax is `start= auto`, not `start=auto`.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- CMD passes text; PowerShell passes objects. Every CMD tool still runs in PowerShell.
- Classify each command as safe, state-changing or destructive before running it; preview and back up the destructive ones.
- Network: ipconfig → ping gateway → nslookup → tracert → port test. Repair: DISM → SFC → chkdsk.
- Never put passwords on a command line: use `net user name *`, `Read-Host -AsSecureString` or `Get-Credential`.
- Scale out with Invoke-Command on the network and Intune Remediations everywhere else.

Check your understanding

1. Why is `Get-Process | Sort-Object WS` more reliable than sorting the output of `tasklist`?
2. A user cannot reach a web app. Ping to the gateway works but `nslookup` fails. What do you check next, and what do you skip?
3. In which order do you run DISM, SFC and chkdsk, and why?
4. A colleague's script creates a local user with the password typed after the user name. Rewrite it in CMD and in PowerShell so the password is never visible.
5. When would you use Intune Remediations instead of `Invoke-Command`?

8.2 Glossary

Cmdlet	A PowerShell command in Verb-Noun form that emits objects.
Elevation	Running with the full administrator token after a UAC prompt.
Component store	`WinSxS`, the master copy of Windows files repaired by DISM.
WinRM	Windows Remote Management, the transport for PowerShell remoting.
SecureString	PowerShell type that keeps a secret encrypted in memory.
Remediation	Intune detection plus remediation script pair.
-WhatIf	Parameter that shows what a cmdlet would change without changing it.